Southern New Hampshire University

CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

This project was started to assist our client, DriverPass, with a new system to enhance their driver training program. The training program will be used as an option to prepare DMV test customers in order to help them achieve a higher success rate on their driving tests.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

The premise behind the DriverPass program is to help prepare drivers so they pass on their first test. To this end DriverPass intends to offer online classes, practice tests and on-site in-car driving instruction.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

Create a customer accounts with all the information needed to make a booking, such as name, street address, phone number, credit card details, as well as pick-up and drop-off locations. The system will allow customers to book an appointment online, choose from one of the three training options available and even change their password if they have forgotten it.

This system will allow access for select staff members, identify which customer is assigned to which driver, vehicle, and time of day for lessons as well as manage bookings. It will also allow removal of training programs, interface with the DMV for updated information and notifications, keep a record of customer's test history and progress which would include test information, time, score and status, and allow the taking of driver's notes including time, start/end time and driver comments.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be *updated?*

- Cloud-based web platform
- Load times for the average user should be in the range of 1-2 seconds
- You should be alerted as to why loading took as long. Aim for a loading time of less than 3 seconds.
- Updates should occur monthly or whenever necessary

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system should be compatible with desktop and mobile versions of browsers including IE, Google Chrome, Firefox and Safari.
- Support for a database to store user information and system details.
- In addition have the backend utilise a web server to manage requests and responses.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- Employees and customers will both have their own login section, employees will be recognised separately.
- Case-sensitive passwords for users
- The system will notify employees daily of any problems

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- All changes relating to users should be done on the backend, you shouldn't need to touch any of the code on the public site.
- Support for platform upgrades.

- IT administrators should be able to access database and web server in order to maintain, update etc. the system.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Requires a password to log into the account.

- Two-factor authentication (2FA) as additional protection for your data and connection.
- Locks an account after 5 failed login attempts.
- "Forgot Password" notify the administrator to change user's password.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- Support for both login username and login password.
- Role support to identify whether user is a customer or admin etc.
- Change your password.
- Lock user out after 5 login attempts with administrator alert.
- Ability to create user accounts and hold user information.
- Online accessibility from any location.
- The ability to show package options to customers.
- Customers can choose and purchase a package.
- Admin can disable fully booked package from control panel
- User assigned to Driver, Schedule, Vehicle must be tracked.
- Automatic DMV database access for the most current license holder information.
- Customer Information, Test Progress and Current Status.
- The system should show driver notes
- The Lesson Timetable should clearly display lessons with start and end times.
- School photos of drivers and students are stored within the program.
- access to online learning tools for customers to utilise as required.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- Access your software online through your favourite web browser on any computer or device.

- The main user groups are: customers and employees / administrators.
- View account details, past orders and contact information.
- A login and registration page
- A home page

- Test scores and progress
- Pass or fail results
- Login details and security settings, including password management and two-step authentication (2FA) configuration, as well as recovery information
- Driver notes, lesson dates, and lesson times
- Package details
- Online learning resources
- A contact information page
- Administrators should be able to view and have access to: Detailed logs of message activity, including successful and unsuccessful mail transfers, recipient addresses, and timestamps.
- Registered customer information
- Package information and corresponding customers.
- ability to remove or disable individual full packages (for embedded images or other split packages)
- Customer and Driver details including names and specific days and times.
- The ability to approve password reset requests

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- Are accessible to all users at any time
- The website should be easy to use.
- administration should be for system administrators, i.e. administrators should be able to handle running the system.
- You are expected to attend all sessions and start on time.
- Be available to teach all assigned classes on time and as scheduled.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- The company has a small fleet of 10 vehicles which required regular servicing.
- The Internet may not be available at times, which may cause delays when downloading updates to this program or watching animations.
- The number of customers able to purchase packs is limited by the number of vehicles.
- Users are using legacy hardware and/or older technology to access the iConnect system.
- The fulfillment of all system requirements may be affected by budget and time constraints.
- Customers may be aware of/require certain types of vehicles and want to rent these out.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

[Insert chart]